

Supplementary material

Legacy standards as frozen solutions to obsolete optimization problems: a framework for technological change and adaptive re-optimization

S1. Pre-specification and frozen selection

The candidate universe contained 122 candidates in 12 domains. 7 roles were frozen before confirmatory estimation. The artifact hash and every input hash are stored in frozen_case_set.json and frozen_case_set.sha256. Expected effect direction, magnitude, significance, and narrative usefulness were excluded from scoring.

S2. Source preservation

Public analysis inputs were saved under data/raw before use. SOURCE_REGISTRY.csv records URL, identifier, version or date, UTC access time, retrieval conditions, local path, file size, SHA-256, terms, completeness, and notes. Existing raw snapshots are never overwritten. A copyrighted ISO 216 sample is retained locally for verification but excluded from the public Git repository.

Table S2. Registered case-specific evidence and data sources.

Case	Source ID	Title	Identifier	SHA-256 prefix
BE03	SEL-BE03-UCI-META	UCI Occupancy Detection dataset metadata	UCI dataset 357; DOI 10.24432/C5X01N	3615bf0faa2a
BE03	SEL-BE03-UCI-DATA	Occupancy Detection data	UCI dataset 357; DOI 10.24432/C5X01N	3648f2aa7026
WK07	SEL-WK07-NASA-DATA	C-MAPSS turbofan engine degradation simulation data	NASA C-MAPSS legacy dataset	74bef434a34d
WK07	SEL-WK07-NASA-PAPER	Review and Analysis of Algorithmic Approaches Developed for Prognostics on CMAPSS Dataset	NASA NTRS 20150007677	5b42232526e3

Case	Source ID	Title	Identifier	SHA-256 prefix
UI01	SEL-UI01-OPSD-DATA	Open Power System Data time series, 60-minute single-index format	OPSD time_series 2020-10-06	6a7f2bc57131
UI01	SEL-UI01-DOE-RATES	Customer Acceptance, Retention, and Response to Time-Based Rates	DOE Consumer Behavior Studies final program impact report	6abbf75f3c09
UI09	SEL-UI09-FAO56	Crop evapotranspiration: Guidelines for computing crop water requirements, revised edition	FAO Irrigation and Drainage Paper 56 Rev.1; ISBN 978-92-5-140060-9	85c5657fa6b8
UI09	SEL-UI09-NASA-POWER	NASA POWER daily agricultural meteorology for Davis, California	NASA POWER point request, Davis CA, 2015-01-01 to 2024-12-31	a8b5e65faf0c
TR01	SEL-TR01-FHWA	Traffic Signal Timing Manual	FHWA-HOP-08-024	675df7afec87
TR01	SEL-TR01-NYC-META	Traffic Volume Counts (Historical) metadata	NYC Open Data btm5-ppia	aee606b1956
TR01	SEL-TR01-NYC-DATA	Traffic Volume Counts (Historical), complete Socrata export	NYC Open Data btm5-ppia	a7ee5895d8d0
NC01	SEL-NC01-ISO668	ISO 668:2020 Series 1 freight containers — Classification, dimensions and ratings	ISO 668:2020	1113d2ed8a76
NC03	SEL-NC03-ISO216	ISO 216:2007 Writing paper and certain classes of printed matter — Trimmed sizes	ISO 216:2007	0fa0831048c2

Case	Source ID	Title	Identifier	SHA-256 prefix
BE03	SEL-BE03-UCI-PAPER	Crossref metadata: Accurate occupancy detection of an office room	10.1016/ j.enbuild.2015.11.071	b1f62383e89c
BE03	SEL-BE03-DOE-SCHEDULES	DOE Commercial Building Energy Asset Score assumptions	Energy Asset Score assumptions document	ffcd5b637bb8
WK07	SEL-WK07-NASA-PBM	Developing Deep Learning Models for System Prognostics	NASA NTRS 20220009583	ab4da4fcc6bb
UI09	SEL-UI09-SENSOR-REVIEW	Crossref metadata: Soil water sensors for irrigation scheduling in the United States	10.1016/ j.agwat.2023.108148	e618a518064e
NC03	AN-NC03-ISO216-SAMPLE	ISO 216:2007 sample pages including the halving and similarity principles	ISO 216:2007 sample 36631	edf66d45765b

S3. Detailed analytical rules

BE03. The UCI CSV is parsed row-wise. Two embedded header rows fail timestamp and occupancy conversion and are removed from the processed analysis table; the raw file is not altered. Fixed schedules are evaluated at 06:00–20:00, 07:00–19:00, and 08:00–18:00. The adaptive policy is an oracle rolling occupancy indicator with the configured hold.

WK07. Training lifetime is the maximum observed cycle for each FD001 training unit. Evaluation lifetime is maximum observed test cycle plus the supplied remaining useful life. A renewal cost rate selects fixed replacement age in the training simulation. The comparator replaces each evaluation unit a configured number of cycles before failure with perfect information.

UI01. Germany-Luxembourg hourly load and day-ahead prices are retained only for complete 24-hour UTC days. Flexible energy is shifted within each day, bounded symmetrically by the

configured fraction of observed hourly load, while daily energy is conserved. The procedure holds prices fixed and is not a retail-tariff response model.

UI09. Daily FAO-56 reference evapotranspiration is calculated from NASA POWER temperature, humidity, wind, radiation, precipitation, latitude, and elevation. Weekly net requirement is non-negative evapotranspiration minus precipitation. The calendar rule uses the training-period median; sensitivity also uses the mean and multiple shortfall penalties.

TR01. The deterministic segment rule selects the segment with the greatest number of days having both NB/SB or EB/WB observations, with segment identifier as the tie-breaker. Training estimates a fixed directional allocation. Evaluation compares it with hourly demand-proportional allocation using a convex saturation-delay proxy. This is not a deployed signal model.

S4. Model sensitivity

Table S1. Pre-specified model sensitivity.

Case	Setting	Estimate	Unit
BE03	fixed_schedule_hours: 08:00-18:00	0.0783925	fraction
BE03	fixed_schedule_hours: 07:00-19:00	0.256686	fraction
BE03	fixed_schedule_hours: 06:00-20:00	0.377177	fraction
WK07	failure_cost_ratio_and_lead: cost=2;lead=1	0.00232759	normalized cost per cycle
WK07	failure_cost_ratio_and_lead: cost=2;lead=5	0.00223097	normalized cost per cycle
WK07	failure_cost_ratio_and_lead: cost=2;lead=10	0.00210467	normalized cost per cycle
WK07	failure_cost_ratio_and_lead: cost=5;lead=1	0.00259603	normalized cost per cycle
WK07	failure_cost_ratio_and_lead: cost=5;lead=5	0.00249941	normalized cost per cycle
WK07	failure_cost_ratio_and_lead: cost=5;lead=10	0.00237311	normalized cost per cycle
WK07	failure_cost_ratio_and_lead: cost=10;lead=1	0.00300736	normalized cost per cycle

Case	Setting	Estimate	Unit
WK07	failure_cost_ratio_and_lead: cost=10;lead=5	0.00291074	normalized cost per cycle
WK07	failure_cost_ratio_and_lead: cost=10;lead=10	0.00278444	normalized cost per cycle
WK07	failure_cost_ratio_and_lead: cost=20;lead=1	0.00300736	normalized cost per cycle
WK07	failure_cost_ratio_and_lead: cost=20;lead=5	0.00291074	normalized cost per cycle
WK07	failure_cost_ratio_and_lead: cost=20;lead=10	0.00278444	normalized cost per cycle
UI01	flexible_load_fraction: 0.005	0.0359227	EUR per MWh baseline load
UI01	flexible_load_fraction: 0.01	0.0718454	EUR per MWh baseline load
UI01	flexible_load_fraction: 0.05	0.359227	EUR per MWh baseline load
UI09	calendar_estimator_and_shor tfall_penalty: median;penalty=2	0.245607	fraction
UI09	calendar_estimator_and_shor tfall_penalty: median;penalty=5	0.456628	fraction
UI09	calendar_estimator_and_shor tfall_penalty: median;penalty=10	0.629402	fraction
UI09	calendar_estimator_and_shor tfall_penalty: mean;penalty=2	0.254153	fraction
UI09	calendar_estimator_and_shor tfall_penalty: mean;penalty=5	0.493914	fraction
UI09	calendar_estimator_and_shor tfall_penalty: mean;penalty=10	0.670467	fraction
TR01	static_green_share: training_estimate	0.272042	fraction
NC03	aspect_ratio: sqrt(2)	4.93038e-32	squared ratio difference

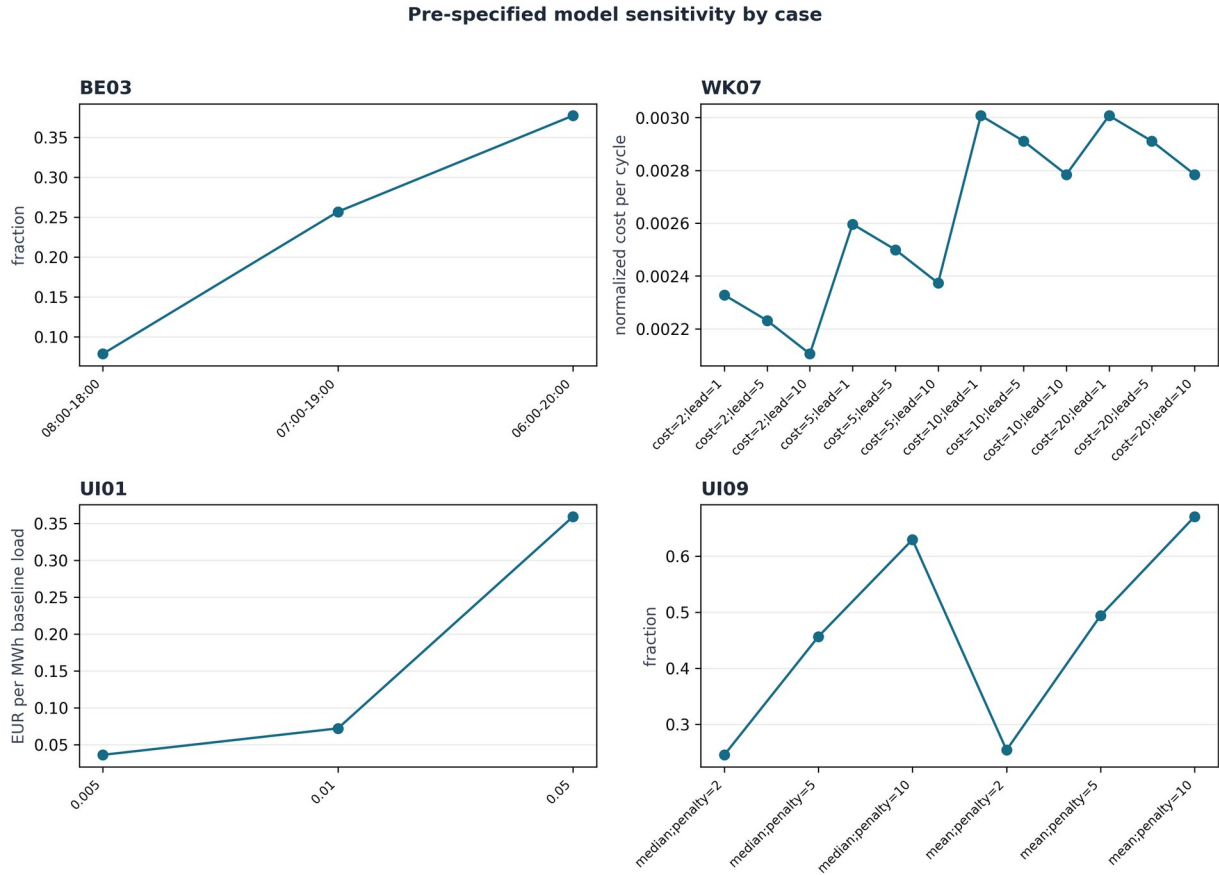


Figure S1. Pre-specified model sensitivity. Each panel uses its own scale.

S5. Cross-case falsification

The analyses do not compute a universal effect size. Leave-one-case-out is also leave-one-domain-out because each primary case represents a distinct domain. A source-quality sensitivity excludes the simulated C-MAPSS case. The negative controls retain the exact repeated-halving null and the unquantified container compatibility case.

Table S3. Full cross-case falsification output.

test	omitted_case	primary_cases_r etained	material_oracle_ mismatches	insufficient_or_u ncertain	interpretation
baseline_primary_ classification_cou nt		4	2	2	Classification count only; incomparable case outcomes are not pooled into a universal effect size.
leave_one_case_a nd_domain_out	BE03	3	2	1	No leave-one-out result licenses a population causal claim; the frozen cases are a designed audit set.
leave_one_case_a nd_domain_out	WK07	3	1	2	No leave-one-out result licenses a population causal claim; the frozen cases are a designed audit set.
leave_one_case_a nd_domain_out	UI01	3	2	1	No leave-one-out result licenses a population causal claim; the frozen cases are a designed audit set.
leave_one_case_a nd_domain_out	UI09	3	1	2	No leave-one-out result licenses a population causal claim; the frozen cases are a designed audit set.
exclude_simulate d_case	WK07	3	1	2	Source-quality sensitivity excluding C- MAPSS simulation.

test	omitted_case	primary_cases_r etained	material_oracle_ mismatches	insufficient_or_u ncertain	interpretation
negative_controls		4	2	2	NC03 returns the pre-specified analytic null; NC01 remains unquantified because transition-cost evidence is absent.

S6. Reproduction

From the project root, create an environment from requirements.lock and run `make all`, `make test`, and `make audit`. Generated case-level processed files are written under outputs/analysis. The manuscript, supplement, figures, tables, and submission archive are regenerated without manual transcription.